

Football Players Market Value Analysis

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Introduction

Portuguese:

Bem vindo ao meu primeiro projeto em Python!

Dataset: "Most Expensive Footballers 2021", disponível no site Kaggle.com

Fonte: <https://www.kaggle.com/datasets/sanjeetsinghnaik/most-expensive-footballers-2021>
(<https://www.kaggle.com/datasets/sanjeetsinghnaik/most-expensive-footballers-2021>)

A análise abaixo foi realizada com o objetivo de entender a relação da métrica "valor de mercado" com as seguintes categorias presentes no dataset:

- Posição
- País
- Idade
- Clube
- Jogadores

Após uma análise geral dessas categorias, criei uma pequena comparação entre os dois jogadores mais valiosos do dataset: Kylian Mbappé e Erling Haaland.

Espero que gostem!

English:

Welcome to my first Python project!

Dataset: "Most Expensive Footballers 2021", available on the website Kaggle.com.

Source: <https://www.kaggle.com/datasets/sanjeetsinghnaik/most-expensive-footballers-2021>
(<https://www.kaggle.com/datasets/sanjeetsinghnaik/most-expensive-footballers-2021>)

The analysis below was conducted to understand the relationship between the market value and the following categories:

- Position
- Country
- Age
- Club
- Players

After that, I compared the two most valuable players of the dataset: Kylian Mbappé and Erling Haaland.

Hope you will enjoy!

1. Loading the libraries and reading the csv

In [1]:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]:  
  
# reading csv  
  
data = pd.read_csv("players.csv")
```

2. Exploratory Analysis

```
In [3]:  
  
data  
  
Out[3]:
```

Unnamed: 0	Name	Position	Age	Markey Value In Millions(£)	Country	Club	Matches	Goals	Own Goals	Assists	Yell Can
0	Kylian Mbappé	Centre-Forward	22	144.0	France	Paris Saint-Germain	16	7	0	11	
1	Erling Haaland	Centre-Forward	21	135.0	Norway	Borussia Dortmund	10	13	0	4	
2	Harry Kane	Centre-Forward	28	108.0	England	Tottenham Hotspur	16	7	0	2	
3	Jack Grealish	Left Winger	26	90.0	England	Manchester City	15	2	0	3	
4	Mohamed Salah	Right Winger	29	90.0	Egypt	Liverpool FC	15	15	0	6	
...	...	...	...	...	...	...	...	...	...	...	...
495	Giorgian de Arrascaeta	Attacking Midfield	27	16.2	Uruguay	Clube de Regatas do Flamengo	0	0	0	0	
496	Ayoze Pérez	Second Striker	28	16.2	Spain	Leicester City	8	1	0	3	
497	Alex Meret	Goalkeeper	24	16.2	Italy	SSC Napoli	5	0	0	0	
498	Duje Caleta-Car	Centre-Back	25	16.2	Croatia	Olympique Marseille	8	0	0	0	
499	Aritz Elustondo	Centre-Back	27	16.2	Spain	Real Sociedad	15	3	0	1	

500 rows × 16 columns

```
In [4]:
display(data.head())
display(data.tail())
```

Unnamed: 0	Name	Position	Age	Markey Value In Millions(£)	Country	Club	Matches	Goals	Own Goals	Assists	Yellow Cards	S
0	0	Kylian Mbappé	Centre-Forward	22	144.0	France	Paris Saint-Germain	16	7	0	11	3
1	1	Erling Haaland	Centre-Forward	21	135.0	Norway	Borussia Dortmund	10	13	0	4	1
2	2	Harry Kane	Centre-Forward	28	108.0	England	Tottenham Hotspur	16	7	0	2	2
3	3	Jack Grealish	Left Winger	26	90.0	England	Manchester City	15	2	0	3	1
4	4	Mohamed Salah	Right Winger	29	90.0	Egypt	Liverpool FC	15	15	0	6	1

Unnamed: 0	Name	Position	Age	Markey Value In Millions(£)	Country	Club	Matches	Goals	Own Goals	Assists	Yellow Card
495	495	Giorgian de Arrascaeta	Attacking Midfield	27	16.2	Uruguay	Clube de Regatas do Flamengo	0	0	0	0
496	496	Ayoze Pérez	Second Striker	28	16.2	Spain	Leicester City	8	1	0	3
497	497	Alex Meret	Goalkeeper	24	16.2	Italy	SSC Napoli	5	0	0	0
498	498	Duje Caleta-Car	Centre-Back	25	16.2	Croatia	Olympique Marseille	8	0	0	0
499	499	Aritz Elustondo	Centre-Back	27	16.2	Spain	Real Sociedad	15	3	0	1

```
In [5]:
# renaming columns
data = data.rename(columns={
    'Markey Value In Millions(£)': 'Market_value',
    'Own Goals': 'Own_goals',
    'Yellow Cards': 'Yellow_cards',
    'Second Yellow Cards': 'Second_yellow_cards',
    'Red Cards': 'Red_cards',
    'Number Of Substitute In': 'Substitute_in',
    'Number Of Substitute Out': 'Substitute_out'})
```

```
In [6]:
# dropping the Unnamed: 0 column
data.drop(columns=['Unnamed: 0'], axis=1, inplace=True)
```

```
In [7]:
data.columns

Out[7]:
Index(['Name', 'Position', 'Age', 'Market_value', 'Country', 'Club', 'Matches',
      'Goals', 'Own_goals', 'Assists', 'Yellow_cards', 'Second_yellow_cards',
      'Red_cards', 'Substitute_in', 'Substitute_out'],
      dtype='object')
```

```
In [8]:
data.index
```

Out[8]:

RangeIndex(start=0, stop=500, step=1)

```
In [9]:
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Name                   500 non-null   object
1   Position               500 non-null   object
2   Age                   500 non-null   int64
3   Market_value           500 non-null   float64
4   Country                500 non-null   object
5   Club                  500 non-null   object
6   Matches               500 non-null   int64
7   Goals                  500 non-null   int64
8   Own_goals              500 non-null   int64
9   Assists                500 non-null   int64
10  Yellow_cards           500 non-null   int64
11  Second_yellow_cards    500 non-null   int64
12  Red_cards              500 non-null   int64
13  Substitute_in          500 non-null   int64
14  Substitute_out         500 non-null   int64
dtypes: float64(1), int64(10), object(4)
memory usage: 58.7+ KB
```

```
In [10]:
data.dtypes
```

Out[10]:

```
Name                object
Position            object
Age                 int64
Market_value        float64
Country             object
Club               object
Matches            int64
Goals              int64
Own_goals          int64
Assists            int64
Yellow_cards       int64
Second_yellow_cards int64
Red_cards          int64
Substitute_in      int64
Substitute_out     int64
dtype: object
```

```
In [11]:
print("We have %.f rows and %.f columns"%(data.shape[0],data.shape[1]))
```

We have 500 rows and 15 columns

In [12]:

data.describe()

Out[12]:

	Age	Market_value	Matches	Goals	Own_goals	Assists	Yellow_cards	Second_yellow_cards
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	24.968000	31.537800	12.396000	2.160000	0.030000	1.512000	1.592000	0.036000
std	3.165916	17.577697	4.342453	2.880102	0.170758	1.85276	1.445585	0.186477
min	16.000000	16.200000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	23.000000	19.800000	10.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	25.000000	25.200000	13.000000	1.000000	0.000000	1.000000	1.000000	0.000000
75%	27.000000	36.000000	16.000000	3.000000	0.000000	2.000000	2.000000	0.000000
max	36.000000	144.000000	24.000000	23.000000	1.000000	12.000000	7.000000	1.000000

In [13]:

data.isnull()

Out[13]:

	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_cards	Secc
0	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...
495	False	False	False	False	False	False	False	False	False	False	False	False
496	False	False	False	False	False	False	False	False	False	False	False	False
497	False	False	False	False	False	False	False	False	False	False	False	False
498	False	False	False	False	False	False	False	False	False	False	False	False
499	False	False	False	False	False	False	False	False	False	False	False	False

500 rows × 15 columns

In [14]:

```
data.isna().sum()
```

the dataset doesn't have missing values

Out[14]:

```
Name                0
Position            0
Age                 0
Market_value        0
Country             0
Club                0
Matches             0
Goals               0
Own_goals            0
Assists              0
Yellow_cards        0
Second_yellow_cards 0
Red_cards           0
Substitute_in       0
Substitute_out      0
dtype: int64
```

In [15]:

```
data['Club'].unique()
```

Out[15]:

```
array(['Paris Saint-Germain', 'Borussia Dortmund', 'Tottenham Hotspur',
      'Manchester City', 'Liverpool FC', 'Chelsea FC',
      'Manchester United', 'FC Barcelona', 'Bayern Munich',
      'Inter Milan', 'Atlético de Madrid', 'West Ham United',
      'Real Sociedad', 'Juventus FC', 'SS Lazio', 'Real Madrid',
      'Bayer 04 Leverkusen', 'Arsenal FC', 'Sevilla FC', 'SSC Napoli',
      'Leicester City', 'Everton FC', 'AC Milan', 'Villarreal CF',
      'ACF Fiorentina', 'RB Leipzig', 'AS Roma', 'Crystal Palace',
      'Wolverhampton Wanderers', 'Valencia CF', 'Leeds United',
      'LOSC Lille', 'Aston Villa', 'Olympique Lyon', 'OGC Nice',
      'AS Monaco', 'Atalanta BC', 'US Sassuolo', 'Torino FC',
      'Ajax Amsterdam', 'Brentford FC', 'Southampton FC',
      'Newcastle United', 'VfL Wolfsburg', 'FC Porto',
      'Olympique Marseille', 'Eintracht Frankfurt',
      'Borussia Mönchengladbach', 'Watford FC', 'Stade Rennais FC',
      'Clube de Regatas do Flamengo', 'VfB Stuttgart', 'Club Brugge KV',
      'Sporting CP', 'Brighton & Hove Albion', 'Dynamo Kyiv',
      'Athletic Bilbao', 'Real Betis Balompí', 'Zenit St. Petersburg',
      'Burnley FC', 'SL Benfica', 'TSG 1899 Hoffenheim', 'Norwich City',
      'PSV Eindhoven', 'KRC Genk', 'Club Atlético Vélez Sarsfield',
      'Club Atlético River Plate', 'FC Metz', 'UC Sampdoria',
      'Red Bull Salzburg', 'Bologna FC 1909', 'Shakhtar Donetsk',
      'Cagliari Calcio', 'Getafe CF', 'Al-Rayyan SC', 'Rubin Kazan',
      'Feyenoord Rotterdam', 'RCD Espanyol Barcelona', 'UD Almería',
      'Sheffield United', 'Celta de Vigo'], dtype=object)
```

In [16]:

```
data.Club.nunique()
```

Out[16]:

81

In [17]:

```
data.Country.unique()
```

Out[17]:

```
array(['France', 'Norway', 'England', 'Egypt', 'Belgium', 'Brazil',  
      'Netherlands', 'Portugal', 'Germany', 'Senegal', 'Korea, South',  
      'Spain', 'Argentina', 'Canada', 'Morocco', 'Italy', 'Serbia',  
      'Slovenia', 'Uruguay', 'Scotland', 'Nigeria', 'Slovakia', 'Poland',  
      "Cote d'Ivoire", 'Austria', 'United States', 'Turkey', 'Mexico',  
      'Croatia', 'Czech Republic', 'Algeria', 'Burkina Faso', 'Sweden',  
      'Ghana', 'Denmark', 'Jamaica', 'Colombia', 'Guinea', 'Switzerland',  
      'Ukraine', 'Russia', 'DR Congo', 'Hungary', 'Mali', 'Japan',  
      'Cameroon', 'Iran', 'Montenegro', 'Gabon', 'Albania', 'Zambia',  
      'The Gambia', 'Israel', 'Georgia', 'Venezuela', 'Wales', 'Peru'],  
      dtype=object)
```

In [18]:

```
data.Country.nunique()
```

Out[18]:

57

In [19]:

```
data.Position.unique()
```

Out[19]:

```
array(['Centre-Forward', 'Left Winger', 'Right Winger',  
      'Attacking Midfield', 'Central Midfield', 'Defensive Midfield',  
      'Right-Back', 'Centre-Back', 'Second Striker', 'Left-Back',  
      'Goalkeeper', 'Left Midfield', 'Right Midfield'], dtype=object)
```

In [20]:

```
data.Position.nunique()
```

Out[20]:

13

```
In [21]:  
  
# Creating new categories for Position using np.where  
  
data['New_Position'] = np.where((data['Position'] == 'Centre-Forward') |  
                                (data['Position'] == 'Left Winger') |  
                                (data['Position'] == 'Right Winger') |  
                                (data['Position'] == 'Second Striker'), 'Attacker',  
                                np.where((data['Position'] == 'Attacking Midfield') |  
                                          (data['Position'] == 'Central Midfield') |  
                                          (data['Position'] == 'Defensive Midfield') |  
                                          (data['Position'] == 'Left Midfield') |  
                                          (data['Position'] == 'Right Midfield'), 'Midfielder',  
                                          np.where(data['Position'] == 'Goalkeeper',  
                                                    'Goalkeeper', 'Defender')))  
  
data
```

Out[21]:

	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_c
0	Kylian Mbappé	Centre-Forward	22	144.0	France	Paris Saint-Germain	16	7	0	11	
1	Erling Haaland	Centre-Forward	21	135.0	Norway	Borussia Dortmund	10	13	0	4	
2	Harry Kane	Centre-Forward	28	108.0	England	Tottenham Hotspur	16	7	0	2	
3	Jack Grealish	Left Winger	26	90.0	England	Manchester City	15	2	0	3	
4	Mohamed Salah	Right Winger	29	90.0	Egypt	Liverpool FC	15	15	0	6	
...	...	...	...	...	...	...	...	...	...	...	
495	Giorgian de Arrascaeta	Attacking Midfield	27	16.2	Uruguay	Clube de Regatas do Flamengo	0	0	0	0	
496	Ayoze Pérez	Second Striker	28	16.2	Spain	Leicester City	8	1	0	3	
497	Alex Meret	Goalkeeper	24	16.2	Italy	SSC Napoli	5	0	0	0	
498	Duje Caleta-Car	Centre-Back	25	16.2	Croatia	Olympique Marseille	8	0	0	0	
499	Aritz Elustondo	Centre-Back	27	16.2	Spain	Real Sociedad	15	3	0	1	

500 rows × 16 columns

```
In [22]:  
  
# converting Market_value from float to int  
  
data['Market_value'] = data['Market_value'].astype(int)  
data.Market_value.dtype  
  
Out[22]:  
  
dtype('int32')
```

```
In [23]:
data

Out[23]:
```

	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_c
0	Kylian Mbappé	Centre-Forward	22	144	France	Paris Saint-Germain	16	7	0	11	
1	Erling Haaland	Centre-Forward	21	135	Norway	Borussia Dortmund	10	13	0	4	
2	Harry Kane	Centre-Forward	28	108	England	Tottenham Hotspur	16	7	0	2	
3	Jack Grealish	Left Winger	26	90	England	Manchester City	15	2	0	3	
4	Mohamed Salah	Right Winger	29	90	Egypt	Liverpool FC	15	15	0	6	
...	...	...	...	...	...	...	...	...	...	...	...
495	Giorgian de Arrascaeta	Attacking Midfield	27	16	Uruguay	Clube de Regatas do Flamengo	0	0	0	0	
496	Ayoze Pérez	Second Striker	28	16	Spain	Leicester City	8	1	0	3	
497	Alex Meret	Goalkeeper	24	16	Italy	SSC Napoli	5	0	0	0	
498	Duje Caleta-Car	Centre-Back	25	16	Croatia	Olympique Marseille	8	0	0	0	
499	Aritz Elustondo	Centre-Back	27	16	Spain	Real Sociedad	15	3	0	1	

500 rows × 16 columns

```
In [24]:  
  
# creating new columns  
  
data['Goals.per.game'] = (data['Goals']/data['Matches']).round(2)  
data['Assists.per.game'] = (data['Assists']/data['Matches']).round(2)  
data
```

Out[24]:

	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_c
0	Kylian Mbappé	Centre-Forward	22	144	France	Paris Saint-Germain	16	7	0	11	
1	Erling Haaland	Centre-Forward	21	135	Norway	Borussia Dortmund	10	13	0	4	
2	Harry Kane	Centre-Forward	28	108	England	Tottenham Hotspur	16	7	0	2	
3	Jack Grealish	Left Winger	26	90	England	Manchester City	15	2	0	3	
4	Mohamed Salah	Right Winger	29	90	Egypt	Liverpool FC	15	15	0	6	
...	...	...	...	...	...	...	...	...	...	...	...
495	Giorgian de Arrascaeta	Attacking Midfield	27	16	Uruguay	Clube de Regatas do Flamengo	0	0	0	0	
496	Ayoze Pérez	Second Striker	28	16	Spain	Leicester City	8	1	0	3	
497	Alex Meret	Goalkeeper	24	16	Italy	SSC Napoli	5	0	0	0	
498	Duje Caleta-Car	Centre-Back	25	16	Croatia	Olympique Marseille	8	0	0	0	
499	Aritz Elustondo	Centre-Back	27	16	Spain	Real Sociedad	15	3	0	1	

500 rows × 18 columns

3. Market Value x Categories

3.1 Position

In [25]:

data

Out[25]:

	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_c
0	Kylian Mbappé	Centre-Forward	22	144	France	Paris Saint-Germain	16	7	0	11	
1	Erling Haaland	Centre-Forward	21	135	Norway	Borussia Dortmund	10	13	0	4	
2	Harry Kane	Centre-Forward	28	108	England	Tottenham Hotspur	16	7	0	2	
3	Jack Grealish	Left Winger	26	90	England	Manchester City	15	2	0	3	
4	Mohamed Salah	Right Winger	29	90	Egypt	Liverpool FC	15	15	0	6	
...	...	...	...	...	...	...	...	...	...	...	...
495	Giorgian de Arrascaeta	Attacking Midfield	27	16	Uruguay	Clube de Regatas do Flamengo	0	0	0	0	
496	Ayoze Pérez	Second Striker	28	16	Spain	Leicester City	8	1	0	3	
497	Alex Meret	Goalkeeper	24	16	Italy	SSC Napoli	5	0	0	0	
498	Duje Caleta-Car	Centre-Back	25	16	Croatia	Olympique Marseille	8	0	0	0	
499	Aritz Elustondo	Centre-Back	27	16	Spain	Real Sociedad	15	3	0	1	

500 rows × 18 columns

In [26]:

data.New_Position.value_counts()

Out[26]:

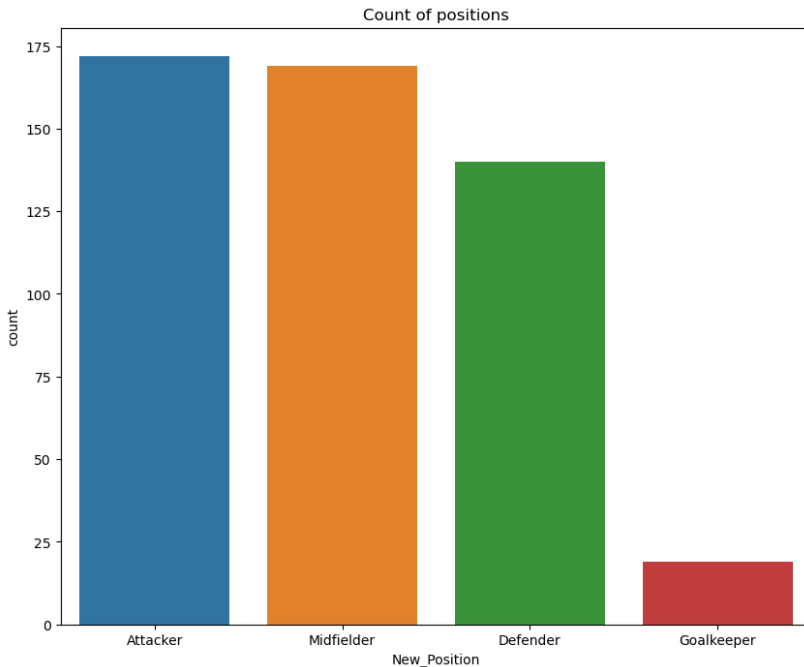
Attacker 172
Midfielder 169
Defender 140
Goalkeeper 19
Name: New_Position, dtype: int64

In [27]:

```
plt.figure(figsize=(10,8))
sns.countplot(data=data, x='New_Position')
plt.title("Count of positions")
```

Out[27]:

Text(0.5, 1.0, 'Count of positions')



In [28]:

```
x = data.groupby('New_Position').Market_value.sum().sort_values(ascending=False).head(10)
x
```

Out[28]:

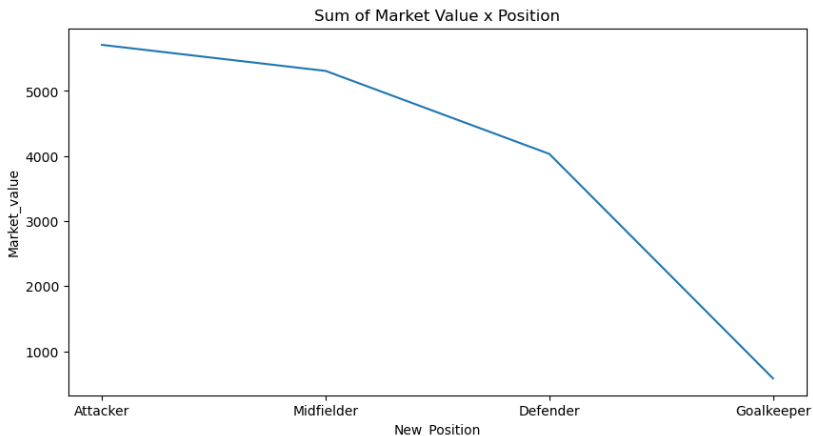
```
New_Position
Attacker      5706
Midfielder    5306
Defender      4031
Goalkeeper     582
Name: Market_value, dtype: int32
```

In [29]:

```
# visualizing
plt.figure(figsize=(10,5))
plt.title("Sum of Market Value x Position")
sns.lineplot(data=x)
```

Out[29]:

```
<Axes: title={'center': 'Sum of Market Value x Position'}, xlabel='New_Position', ylabel='Market_value'>
```



In [30]:

```
y = data.groupby('New_Position').Market_value.mean().sort_values(ascending=False).head(10)
y
```

Out[30]:

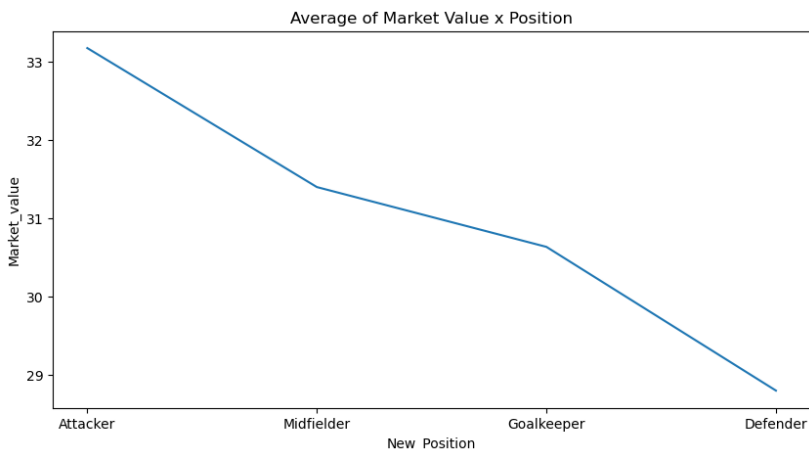
```
New_Position
Attacker      33.174419
Midfielder    31.396450
Goalkeeper    30.631579
Defender      28.792857
Name: Market_value, dtype: float64
```

In [31]:

```
# visualizing
plt.figure(figsize=(10,5))
plt.title("Average of Market Value x Position")
sns.lineplot(data=y)
```

Out[31]:

```
<Axes: title={'center': 'Average of Market Value x Position'}, xlabel='New_Position', ylabel='Market_value'>
```



In [32]:

```
# which are the specific positions of the field that are most valued?
```

```
data.Position.value_counts()
```

Out[32]:

```
Centre-Back      87
Central Midfield  74
Centre-Forward   70
Right Winger     48
Left Winger      46
Attacking Midfield 41
Defensive Midfield 41
Right-Back       30
Left-Back        23
Goalkeeper       19
Second Striker    8
Left Midfield     8
Right Midfield    5
Name: Position, dtype: int64
```

```
In [33]:
c=data.groupby('Position').Market_value.sum().sort_values(ascending = False)
c
```

Out[33]:

Position	
Centre-Back	2557
Central Midfield	2401
Centre-Forward	2348
Left Winger	1632
Right Winger	1447
Attacking Midfield	1324
Defensive Midfield	1263
Right-Back	777
Left-Back	697
Goalkeeper	582
Second Striker	279
Left Midfield	213
Right Midfield	105

Name: Market_value, dtype: int32

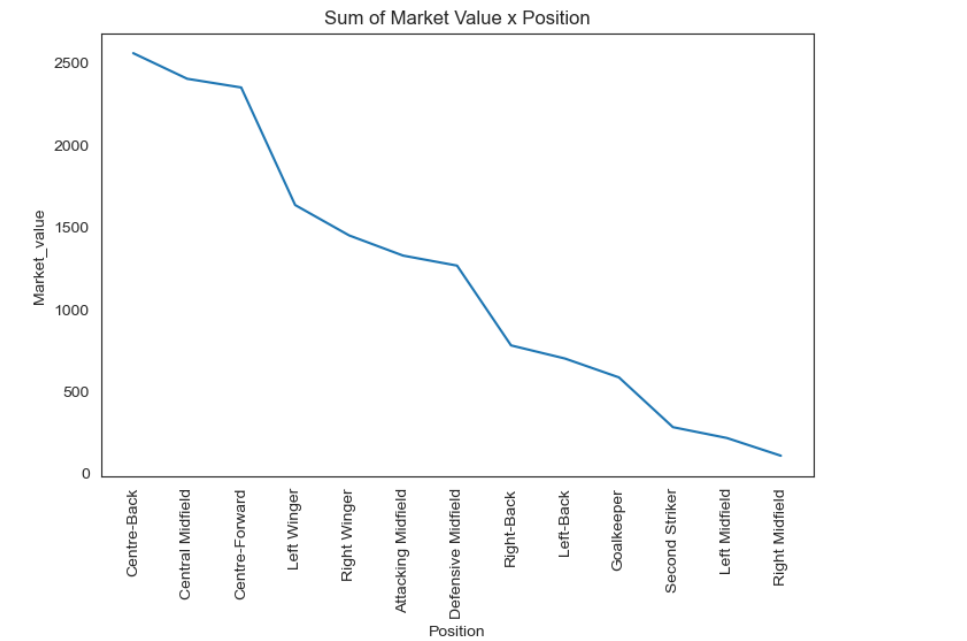
```
In [34]:
data.groupby('Position').Market_value.agg(['max', 'min']).sort_values(by=['max', 'min'], ascending = False)
```

Out[34]:

	max	min
Position		
Centre-Forward	144	16
Attacking Midfield	90	16
Left Winger	90	16
Right Winger	90	16
Central Midfield	81	16
Defensive Midfield	81	16
Centre-Back	67	16
Right-Back	67	16
Goalkeeper	63	16
Left-Back	63	16

```
In [70]:  
  
# visualizing  
  
plt.figure(figsize=(8,5))  
  
plt.title("Sum of Market Value x Position")  
  
plt.xticks(rotation=90)  
  
sns.lineplot(data=c)
```

Out[70]:
<Axes: title={'center': 'Sum of Market Value x Position'}, xlabel='Position', ylabel='Market_value'>



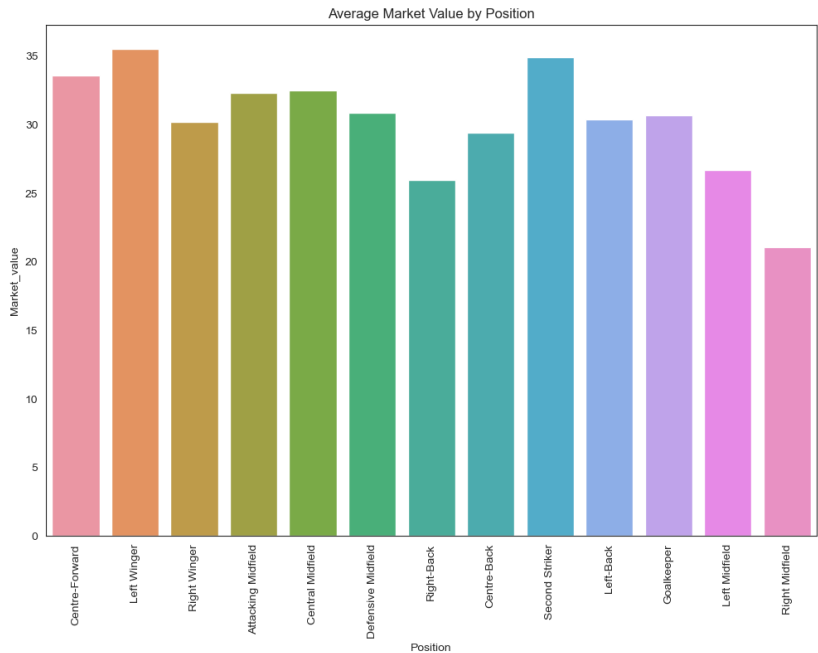
```
In [36]:  
  
data.groupby("Position").Market_value.mean().sort_values(ascending=False)
```

Out[36]:

Position	
Left Winger	35.478261
Second Striker	34.875000
Centre-Forward	33.542857
Central Midfield	32.445946
Attacking Midfield	32.292683
Defensive Midfield	30.804878
Goalkeeper	30.631579
Left-Back	30.304348
Right Winger	30.145833
Centre-Back	29.390805
Left Midfield	26.625000
Right-Back	25.900000
Right Midfield	21.000000

Name: Market_value, dtype: float64

```
In [72]:
plt.figure(figsize=(12, 8))
sns.barplot(data=data, x='Position', y='Market_value',errorbar=None)
plt.title('Average Market Value by Position')
sns.set_style("white")
plt.xticks(rotation=90)
plt.show()
```



Insights

Portuguese

- Analisando pelas posições gerais, notamos que o dataset tem maior número de atacantes, seguido de meiocampistas e defensores. Os goleiros aparecem em menor quantidade.
- Notamos também que os atacantes são mais valorizados, seguidos dos meias e os defensores. Os goleiros tem uma soma de valor de mercado bem inferior. Porém, quando observamos a média, os defensores ficam abaixo dos goleiros.
- Analisando as posições específicas (que estão dentro de atacantes, meiocampistas e defensores), notamos que temos um grande número de zagueiros, meiocampistas centrais e centroavantes no dataset. Por conta da maior quantidade, essas posições também apresentaram a maior soma de valor de mercado.
- Porém, quando observamos a média, pontas-esquerdos, segundos-atacantes e centroavantes possuem o maior índice.

English

- Analyzing by general positions, we notice that the dataset has a higher number of forwards, followed by midfielders and defenders. The goalkeepers appear in a smaller quantity.
- We can also observe that forwards are more valued, followed by midfielders and defenders. Goalkeepers have a significantly lower total market value. However, when we look at the average, defenders are below goalkeepers.
- Analyzing specific positions within forwards, midfielders, and defenders, we notice that we have a large number of center-backs, central midfielders, and center-forwards in the dataset. Due to the higher quantity, these positions also have the highest sum of market value.
- However, when we consider the average, left-wingers, second strikers, and center-forwards have the highest value index.

3.2 Country

```
In [38]:
data.Country.value_counts().head(10)

Out[38]:
England      67
France       58
Spain        52
Brazil       41
Germany      29
Portugal     26
Italy        26
Argentina    22
Netherlands  17
Belgium      14
Name: Country, dtype: int64

In [39]:
data.groupby('Country').Market_value.sum().sort_values(ascending=False).head(10)

Out[39]:
Country
England      2227
France       1878
Spain        1556
Brazil       1263
Germany       999
Portugal      881
Italy         847
Argentina     645
Netherlands   565
Belgium       519
Name: Market_value, dtype: int32

In [40]:
data.groupby('Country').Market_value.agg(['max', 'min']).sort_values(by=['max', 'min'], ascending = False).

Out[40]:
```

	max	min
Country		
France	144	16
Norway	135	16
England	108	16
Egypt	90	90
Belgium	90	16
Brazil	90	16
Netherlands	81	19
Germany	81	16
Portugal	81	16
Korea, South	76	76

```
In [41]:
data.groupby('Country').Market_value.mean().sort_values(ascending = False).head(10)
```

Out[41]:

Country	Market_value
Egypt	90.000000
Korea, South	76.000000
Slovenia	63.000000
Norway	62.333333
Slovakia	54.000000
Canada	47.000000
Morocco	37.250000
Belgium	37.071429
Ghana	36.000000
Senegal	35.000000

Name: Market_value, dtype: float64

```
In [42]:
selected_countries = ['Slovenia', 'Norway', 'Slovakia', 'Canada', 'Morocco', 'Belgium', 'Ghana', 'Senegal']
counts = data[data['Country'].isin(selected_countries)]['Country'].value_counts()
counts
```

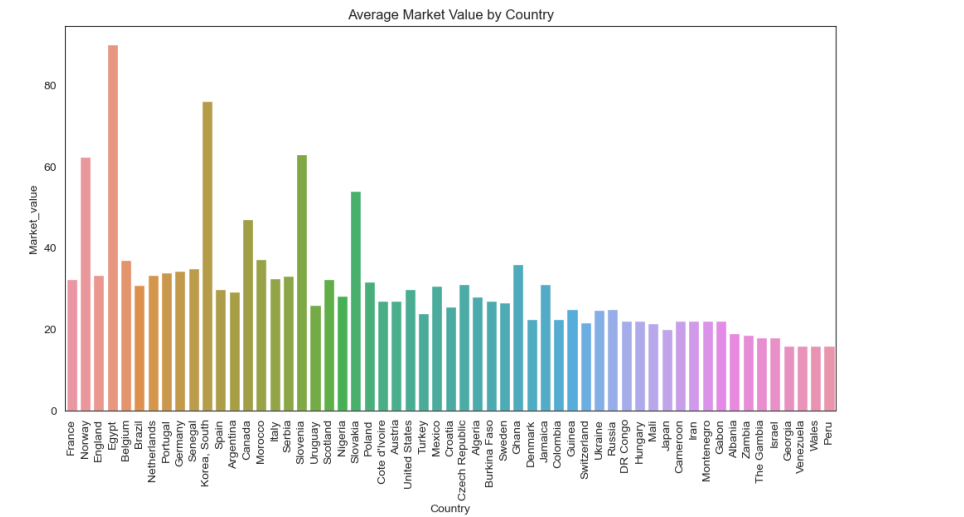
Out[42]:

Country	count
Belgium	14
Senegal	6
Norway	3
Canada	2
Slovenia	1
Slovakia	1
Ghana	1

Name: Country, dtype: int64

```
In [78]:
# visualizing

plt.figure(figsize=(12, 6))
sns.barplot(data=data, x='Country', y='Market_value', errorbar=None)
plt.title('Average Market Value by Country')
plt.xticks(rotation=90)
plt.show()
```



Insights

Portuguese

- Inglaterra tem o maior número de jogadores no dataset, seguido de França, Espanha, Brasil e Alemanha. Esses mesmos cinco países tem as maiores somas de valor de mercado, por conta de possuírem mais jogadores na lista.
- França e Noruega possuem o maior valor máximo, por conta de terem os dois jogadores mais valorizados. O Egito possui o mesmo valor de máximo e mínimo, além do maior valor médio. Isso se da pelo país ter apenas um jogador na lista (Salah).
- Outro países apresentaram média elevada por conta de terem poucos jogadores na base de dados, como a Coreia do Sul, Noruega, Slovakia e Slovenia.

English

- England has the highest number of players in the dataset, followed by France, Spain, Brazil, and Germany. These same five countries have the highest total market values due to having more players on the list.
- France and Norway have the highest maximum value, as they have the two most valued players. Egypt has the same maximum and minimum value, along with the highest average value, because the country has only one player in the list (Salah).
- Other countries have higher averages because they have few players in the database, such as South Korea, Norway, Slovakia, and Slovenia.

3.3 Age

In [44]:

```
data.Age.value_counts()
```

Out[44]:

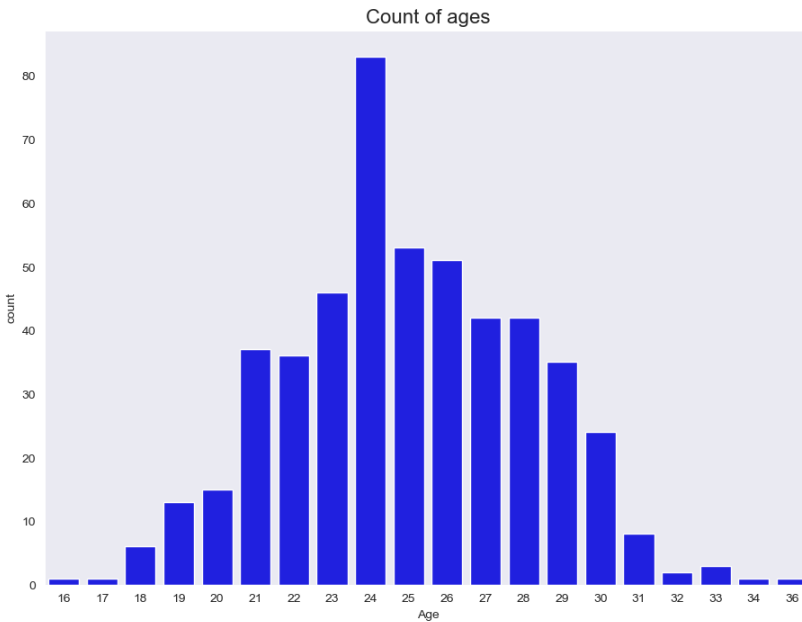
```
24    83
25    53
26    51
23    46
27    42
28    42
21    37
22    36
29    35
30    24
20    15
19    13
31     8
18     6
33     3
32     2
34     1
36     1
17     1
16     1
Name: Age, dtype: int64
```

In [45]:

```
plt.figure(figsize=(11,8))  
sns.set_style("dark")  
plt.title("Count of ages", fontsize=16)  
plt.xlabel("Age")  
sns.countplot(data=data, x="Age", color='blue')
```

Out[45]:

<Axes: title={'center': 'Count of ages'}, xlabel='Age', ylabel='count'>



```
In [79]:
plt.figure(figsize=(12, 8))
sns.distplot(data['Age'], hist=True, color='Blue')
plt.title("Density Plot of the Ages", fontsize=16)
```

C:\Users\guilh\AppData\Local\Temp\ipykernel_12256\1149527092.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

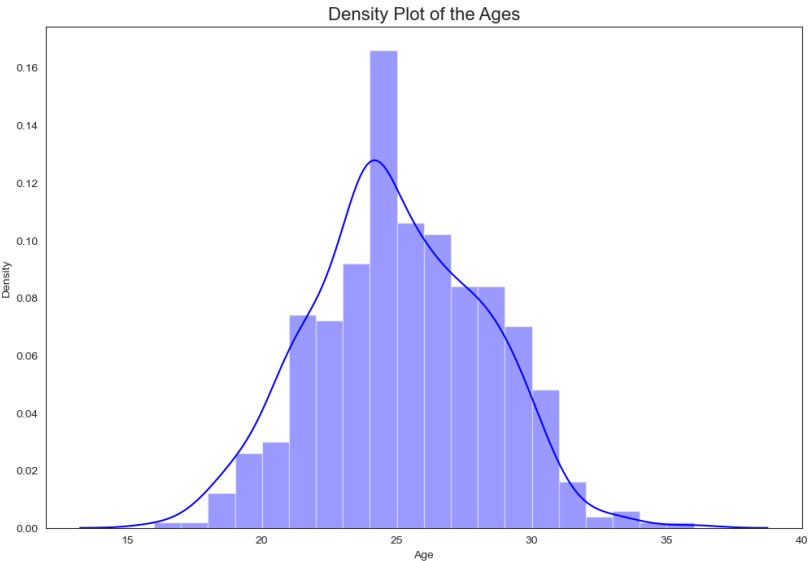
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751> (<https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>)

```
sns.distplot(data['Age'], hist=True, color='Blue')
```

Out[79]:

Text(0.5, 1.0, 'Density Plot of the Ages')



```
In [47]:
# age x market value sum
data.groupby('Age').Market_value.sum().sort_values(ascending=False).head(10)
```

Out[47]:

Age	
24	2526
26	1694
25	1409
28	1380
29	1350
22	1275
23	1271
27	1261
21	1185
30	729

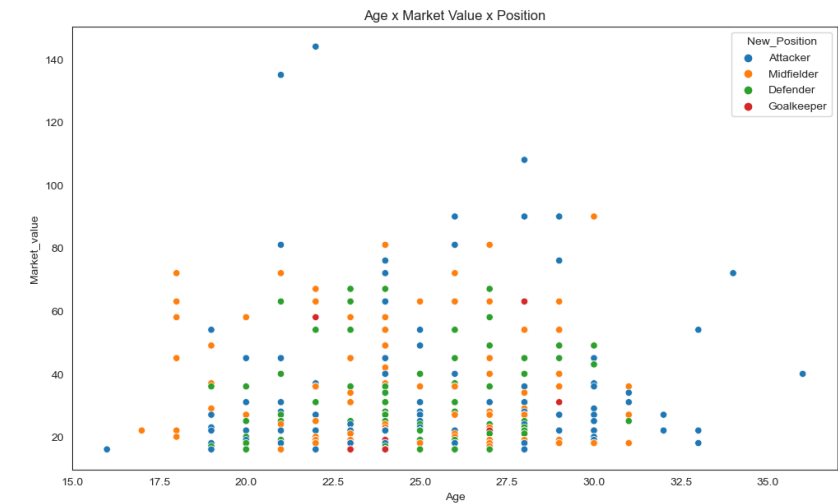
Name: Market_value, dtype: int32

```
In [48]:  
  
# age x market value average  
  
data.groupby('Age').Market_value.mean().sort_values(ascending=False).head(10)
```

Out[48]:

Age
34 72.000000
18 46.666667
36 40.000000
29 38.571429
22 35.416667
26 33.215686
28 32.857143
21 32.027027
33 31.333333
24 30.433735
Name: Market_value, dtype: float64

```
In [80]:  
  
plt.figure(figsize=(12,7))  
  
plt.title("Age x Market Value x Position")  
  
sns.scatterplot(data=data, x='Age', y='Market_value', hue='New_Position')  
  
sns.set_style("darkgrid")
```



Insights

Portuguese

- A maior parte dos jogadores do dataset estão na faixa dos 20 até 30 anos.
- Temos a maior quantidade de jogadores com 24 anos(83 jogadores), seguido dos 25 anos (53 jogadores) e 26 anos (51 jogadores). Por conta da maior volumetria de jogadores com essas idades, elas concentraram maior soma de valor de mercado.

English

- Most of the players in the dataset are in the age range of 20 to 30 years.
- We have a higher number of players aged 24 (83 players), followed by 25 years (53 players), and 26 years (51 players). Due to the larger number of players in these age groups, they have accumulated a higher total market value.

3.4 Clubs

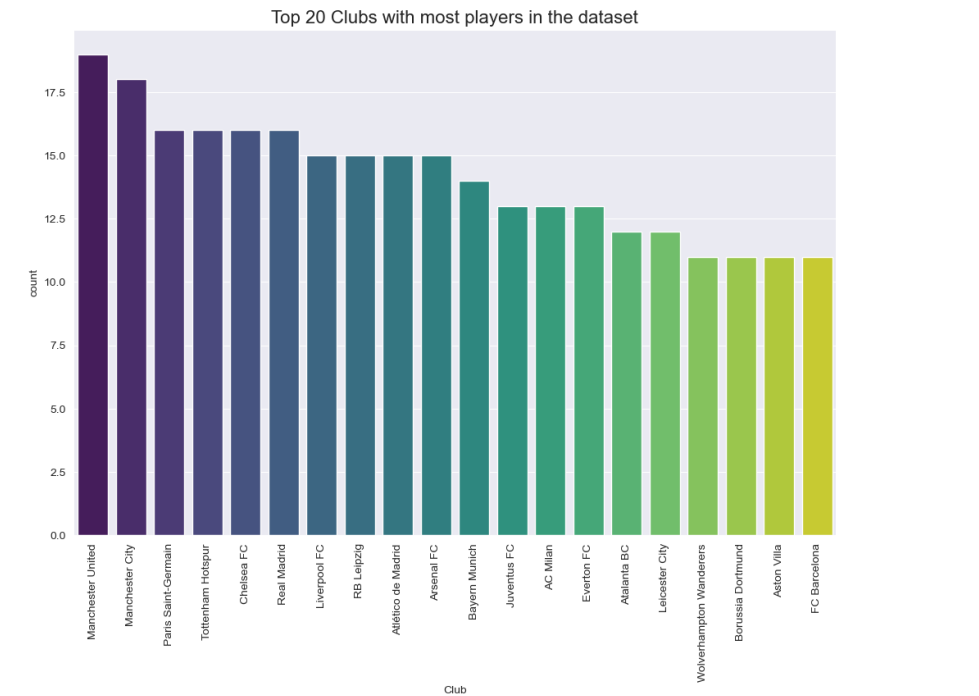
```
In [50]:
data.Club.unique()

Out[50]:
array(['Paris Saint-Germain', 'Borussia Dortmund', 'Tottenham Hotspur',
      'Manchester City', 'Liverpool FC', 'Chelsea FC',
      'Manchester United', 'FC Barcelona', 'Bayern Munich',
      'Inter Milan', 'Atlético de Madrid', 'West Ham United',
      'Real Sociedad', 'Juventus FC', 'SS Lazio', 'Real Madrid',
      'Bayer 04 Leverkusen', 'Arsenal FC', 'Sevilla FC', 'SSC Napoli',
      'Leicester City', 'Everton FC', 'AC Milan', 'Villarreal CF',
      'ACF Fiorentina', 'RB Leipzig', 'AS Roma', 'Crystal Palace',
      'Wolverhampton Wanderers', 'Valencia CF', 'Leeds United',
      'LOSC Lille', 'Aston Villa', 'Olympique Lyon', 'OGC Nice',
      'AS Monaco', 'Atalanta BC', 'US Sassuolo', 'Torino FC',
      'Ajax Amsterdam', 'Brentford FC', 'Southampton FC',
      'Newcastle United', 'VFL Wolfsburg', 'FC Porto',
      'Olympique Marseille', 'Eintracht Frankfurt',
      'Borussia Mönchengladbach', 'Watford FC', 'Stade Rennais FC',
      'Clube de Regatas do Flamengo', 'VfB Stuttgart', 'Club Brugge KV',
      'Sporting CP', 'Brighton & Hove Albion', 'Dynamo Kyiv',
      'Athletic Bilbao', 'Real Betis Balompie', 'Zenit St. Petersburg',
      'Burnley FC', 'SL Benfica', 'TSG 1899 Hoffenheim', 'Norwich City',
      'PSV Eindhoven', 'KRC Genk', 'Club Atlético Vélez Sarsfield',
      'Club Atlético River Plate', 'FC Metz', 'UC Sampdoria',
      'Red Bull Salzburg', 'Bologna FC 1909', 'Shakhtar Donetsk',
      'Cagliari Calcio', 'Getafe CF', 'Al-Rayyan SC', 'Rubin Kazan',
      'Feyenoord Rotterdam', 'RCD Espanyol Barcelona', 'UD Almeria',
      'Sheffield United', 'Celta de Vigo'], dtype=object)

In [51]:
top_20_clubs = data.Club.value_counts().head(20)
top_20_clubs

Out[51]:
Manchester United      19
Manchester City        18
Paris Saint-Germain    16
Tottenham Hotspur     16
Chelsea FC             16
Real Madrid            16
Liverpool FC           15
RB Leipzig             15
Atlético de Madrid    15
Arsenal FC             15
Bayern Munich          14
Juventus FC            13
AC Milan               13
Everton FC             13
Atalanta BC          12
Leicester City         12
Wolverhampton Wanderers 11
Borussia Dortmund      11
Aston Villa            11
FC Barcelona           11
Name: Club, dtype: int64
```

```
In [52]:  
  
# visualizing  
  
plt.figure(figsize=(12, 8))  
  
plt.title("Top 20 Clubs with most players in the dataset", fontsize=16)  
  
sns.countplot(data=data, x='Club', order=top_20_clubs.index, palette='viridis')  
  
plt.xticks(rotation=90)  
  
plt.show()
```



In [53]:

```
data.groupby('Club').Market_value.sum().sort_values(ascending = False).nlargest(20)
```

Out[53]:

Club	
Manchester City	937
Paris Saint-Germain	772
Manchester United	754
Chelsea FC	704
Bayern Munich	683
Liverpool FC	677
Atlético de Madrid	615
Real Madrid	590
Tottenham Hotspur	530
FC Barcelona	448
Juventus FC	447
Borussia Dortmund	422
Arsenal FC	403
Leicester City	385
RB Leipzig	375
Inter Milan	373
AC Milan	354
SSC Napoli	351
Everton FC	316
Aston Villa	284

Name: Market_value, dtype: int32

In [54]:

```
top_20_clubs_value = data.groupby('Club').Market_value.mean().sort_values(ascending = False).nlargest(20)  
top_20_clubs_value
```

Out[54]:

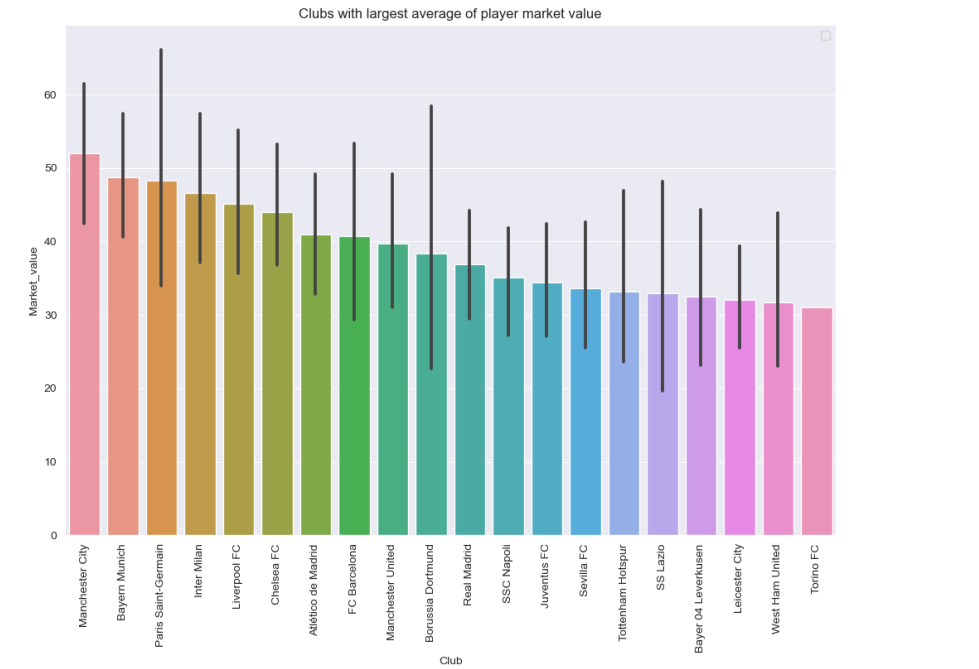
Club	
Manchester City	52.055556
Bayern Munich	48.785714
Paris Saint-Germain	48.250000
Inter Milan	46.625000
Liverpool FC	45.133333
Chelsea FC	44.000000
Atlético de Madrid	41.000000
FC Barcelona	40.727273
Manchester United	39.684211
Borussia Dortmund	38.363636
Real Madrid	36.875000
SSC Napoli	35.100000
Juventus FC	34.384615
Sevilla FC	33.666667
Tottenham Hotspur	33.125000
SS Lazio	33.000000
Bayer 04 Leverkusen	32.500000
Leicester City	32.083333
West Ham United	31.714286
Torino FC	31.000000

Name: Market_value, dtype: float64

```
In [55]:  
  
# visualizing  
  
plt.figure(figsize=(12,8))  
  
plt.title("Clubs with largest average of player market value")  
  
sns.barplot(data=data, x='Club', y='Market_value', order=top_20_clubs_value.index)  
  
sns.set_style("dark")  
  
plt.xticks(rotation=90)  
  
plt.legend()
```

No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

```
Out[55]:  
  
<matplotlib.legend.Legend at 0x26578ce0850>
```



Insights

Portuguese

- Conforme vimos, temos múltiplos clubes no dataset. Os que possuem a maior quantidade de jogadores são os seguintes: Manchester United (19 jogadores), Manchester City (18 jogadores), Paris Saint-Germain (16 jogadores), Tottenham Hotspur (16 jogadores), Chelsea FC (16 jogadores) e Real Madrid (16 jogadores).
- O Manchester City lidera na soma de valor de mercado, seguido do Paris Saint-Germain, Manchester United, Chelsea FC e Bayern Munich.
- Quando observamos a média de valor de mercado por clube, notamos que o Manchester City, Bayern de Munique, Paris Saint-Germain, Inter de Milão e Liverpool possuem os maiores valores, o que indica o quão valiosos são os jogadores dessas equipes.

English

- As we have seen, there are multiple clubs in the dataset. The ones with the highest number of players are as follows: Manchester United (19 players), Manchester City (18 players), Paris Saint-Germain (16 players), Tottenham Hotspur (16 players), Chelsea FC (16 players), and Real Madrid (16 players).
- Manchester City leads in the total market value sum, followed by Paris Saint-Germain, Manchester United, Chelsea FC, and Bayern Munich.
- When we observe the average market value per club, we notice that Manchester City, Bayern Munich, Paris Saint-Germain, Inter Milan, and Liverpool have the highest values, indicating how valuable the players of these teams are.

3.5 Players

```
In [56]:
data.Name.unique()

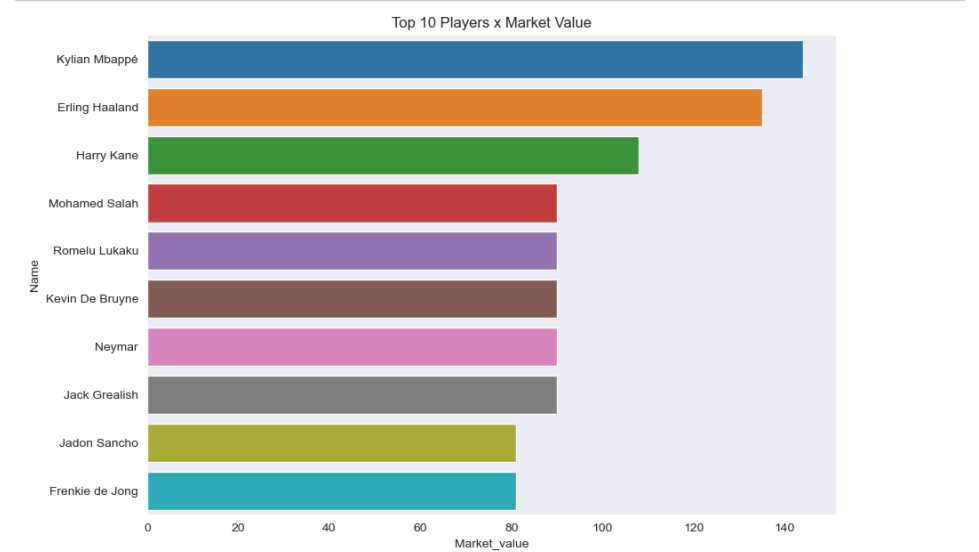
Out[56]:
array(['Kylian Mbappé', 'Erling Haaland', 'Harry Kane', 'Jack Grealish',
      'Mohamed Salah', 'Romelu Lukaku', 'Kevin De Bruyne', 'Neymar',
      'Jadon Sancho', 'Frenkie de Jong', 'Bruno Fernandes',
      'Joshua Kimmich', 'Raheem Sterling', 'Marcus Rashford',
      'Sadio Mané', 'Heung-min Son', 'Pedri', 'Phil Foden',
      'Lautaro Martinez', 'Marcos Llorente', 'Lionel Messi',
      'Mason Mount', 'Trent Alexander-Arnold', 'Rúben Dias',
      'Marquinhos', 'Jude Bellingham', 'João Félix', 'Alphonso Davies',
      'Achraf Hakimi', 'Declan Rice', 'Rodri', 'Mikel Oyarzabal',
      'Federico Chiesa', 'Matthijs de Ligt', 'Kai Havertz',
      'Sergej Milinković-Savić', 'Bernardo Silva', 'Raphaël Varane',
      'Serge Gnabry', 'Leon Goretzka', 'Jan Oblak', 'Casemiro',
      'Florian Wirtz', 'Bukayo Saka', 'Federico Valverde',
      'Gianluigi Donnarumma', 'Nicolò Barella', 'Andrew Robertson',
      'Ansu Fati', 'Jules Koundé', 'Victor Osimhen', 'Gabriel Jesus',
      'Dayot Upamecano', 'Alessandro Bastoni', 'Wilfred Ndidi',
      'José María Giménez', 'Fabinho', 'Milan Skriniar', 'Leroy Sané',
      'Paul Pogba', 'Thibaut Courtois', 'Alisson', 'Koke']
```

```
In [57]:  
  
# analyzing the top 10 players  
  
top_10 = data.sort_values(ascending = False, by='Market_value').head(10)  
top_10
```

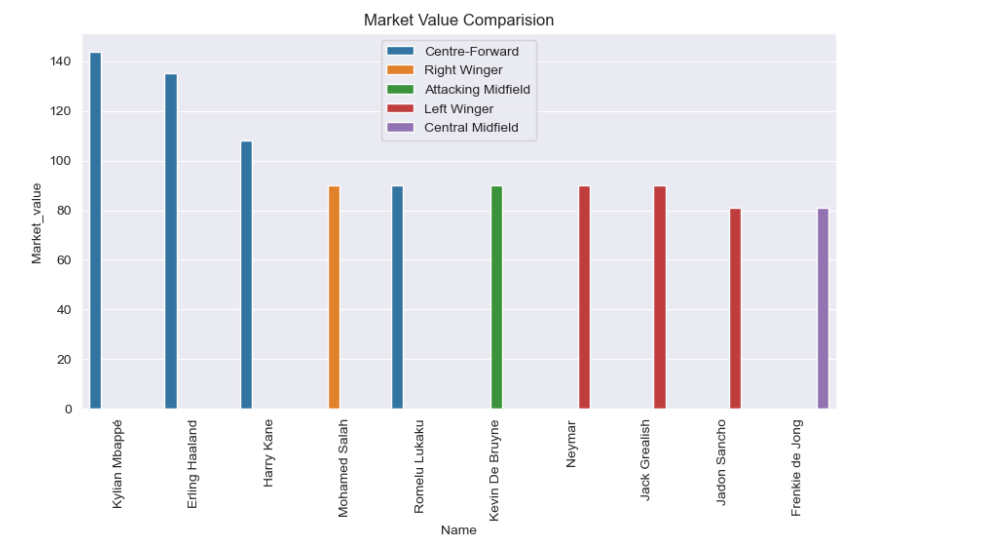
Out[57]:

	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_cards
0	Kylian Mbappé	Centre-Forward	22	144	France	Paris Saint-Germain	16	7	0	11	
1	Erling Haaland	Centre-Forward	21	135	Norway	Borussia Dortmund	10	13	0	4	
2	Harry Kane	Centre-Forward	28	108	England	Tottenham Hotspur	16	7	0	2	
4	Mohamed Salah	Right Winger	29	90	Egypt	Liverpool FC	15	15	0	6	
5	Romelu Lukaku	Centre-Forward	28	90	Belgium	Chelsea FC	11	4	0	1	
6	Kevin De Bruyne	Attacking Midfield	30	90	Belgium	Manchester City	14	3	0	1	
7	Neymar	Left Winger	29	90	Brazil	Paris Saint-Germain	11	3	0	3	
3	Jack Grealish	Left Winger	26	90	England	Manchester City	15	2	0	3	
8	Jadon Sancho	Left Winger	21	81	England	Manchester United	13	0	0	0	
9	Frenkie de Jong	Central Midfield	24	81	Netherlands	FC Barcelona	13	0	0	2	

```
In [58]:  
  
#visualizing  
  
plt.figure(figsize=(10,7))  
  
plt.title("Top 10 Players x Market Value")  
  
sns.barplot(data=top_10, y="Name", x="Market_value")  
  
sns.set_style("darkgrid")
```



```
In [81]:  
  
# visualizing  
  
plt.figure(figsize=(10,5))  
  
plt.title("Market Value Comparision")  
  
sns.barplot(x='Name', y='Market_value', hue='Position', data=top_10)  
  
plt.xticks(rotation=90)  
  
plt.legend()  
  
Out[81]:  
  
<matplotlib.legend.Legend at 0x2657e83bc40>
```



```
In [60]:
plt.figure(figsize=(10,8))

plt.title("Players x Position x Market Value")

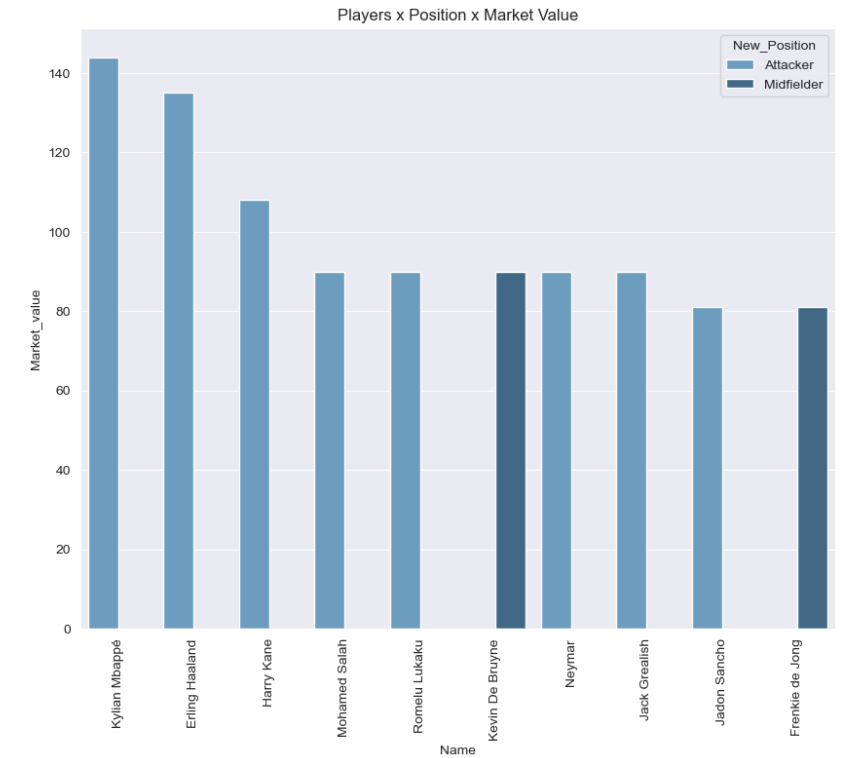
plt.xticks(rotation=90)

sns.set_style("dark")

sns.barplot(x='Name', y='Market_value', hue='New_Position', data=top_10, palette='Blues_d')
```

Out[60]:

<Axes: title={'center': 'Players x Position x Market Value'}, xlabel='Name', ylabel='Market_value'>



In [61]:

```
data.groupby(['Name', 'Club']).Market_value.sum().sort_values(ascending = False).nlargest(10)
```

Out[61]:

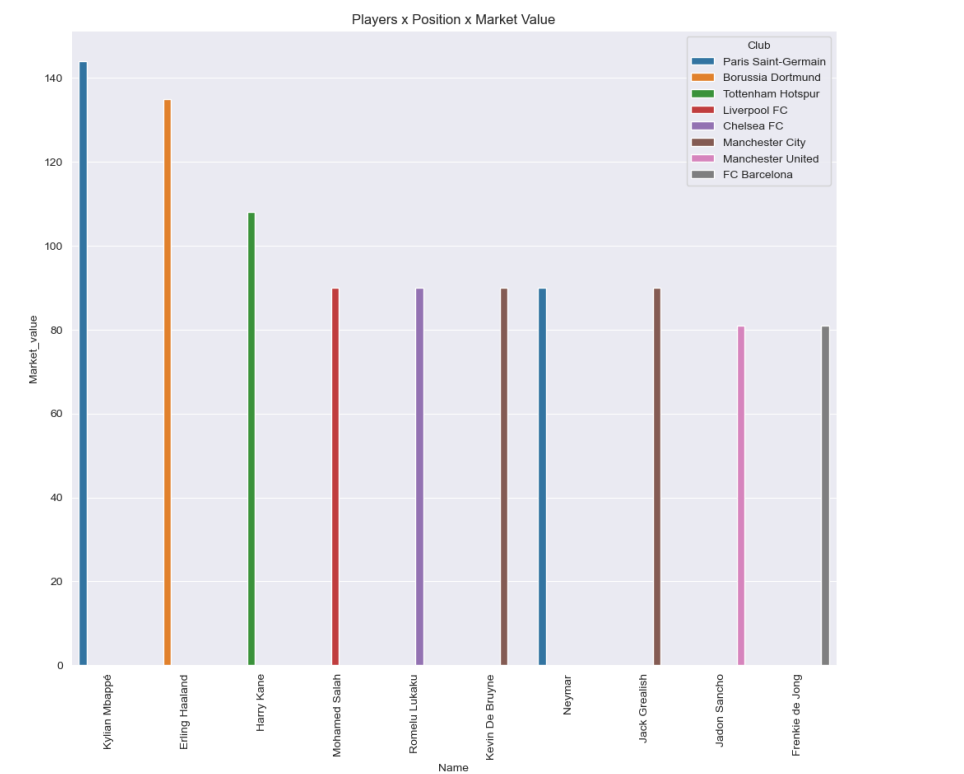
Name	Club	
Kylian Mbappé	Paris Saint-Germain	144
Erling Haaland	Borussia Dortmund	135
Harry Kane	Tottenham Hotspur	108
Neymar	Paris Saint-Germain	90
Mohamed Salah	Liverpool FC	90
Kevin De Bruyne	Manchester City	90
Romelu Lukaku	Chelsea FC	90
Jack Grealish	Manchester City	90
Bruno Fernandes	Manchester United	81
Raheem Sterling	Manchester City	81

Name: Market_value, dtype: int32

```
In [82]:  
  
# visualizing  
  
plt.figure(figsize=(12,10))  
  
plt.title("Players x Position x Market Value")  
  
plt.xticks(rotation=90)  
  
sns.set_style("dark")  
  
sns.barplot(x='Name', y='Market_value', hue='Club', data=top_10)
```

Out[82]:

<Axes: title={'center': 'Players x Position x Market Value'}, xlabel='Name', ylabel='Market_v
alue'>



Insights

Portuguese

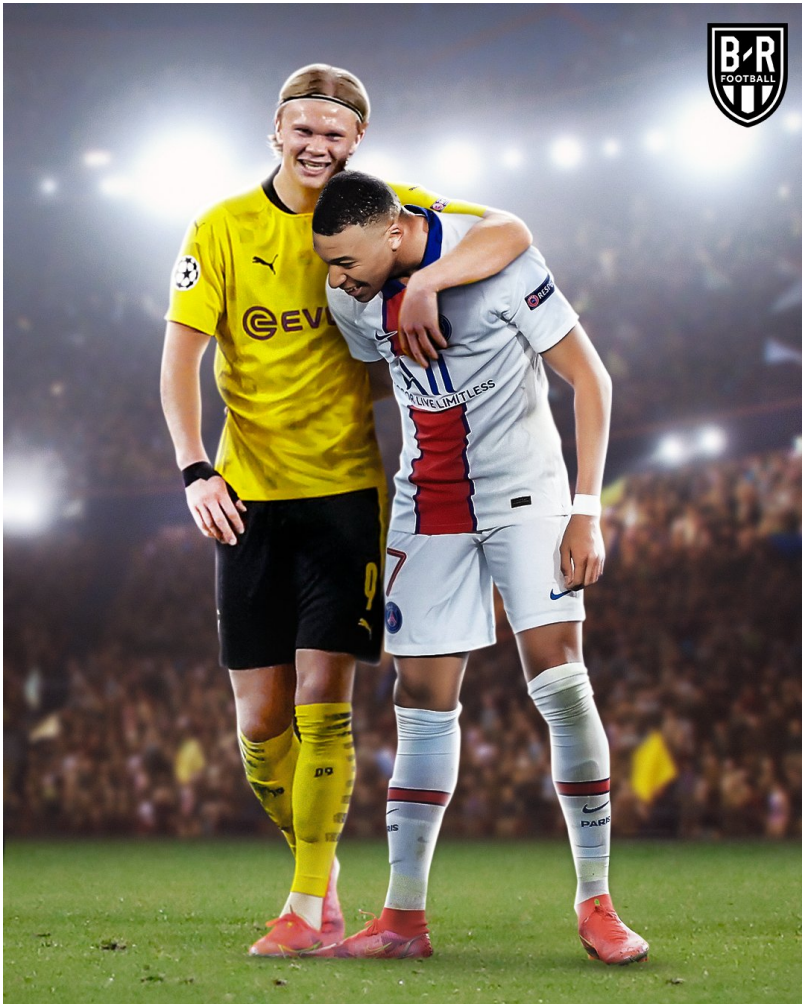
- Análise os 10 jogadores com maior valor de mercado do dataset. Como podemos observar, Kylian Mbappé, Erling Haaland, Harry Kane, Mohamed Salah e Lukaku lideram a lista.
- Dentre esses 10 jogadores, 8 são atacantes, enquanto 2 são meiocampistas. Além disso, temos 4 Centre-Forwards, 3 Left-Wingers, 1 Right-Winger, 1 Attacking Midfield e 1 Center-Midfielder.

- Analisando por clube, temos dois jogadores do Manchester City e dois do Paris Saint-Germain entre os 10 mais valorizados.

English

- I analyzed the top 10 players with the highest market value in the dataset. As we can see, Kylian Mbappé, Erling Haaland, Harry Kane, Mohamed Salah, and Lukaku are leading the list.
- Among these 10 players, 8 are forwards, while 2 are midfielders. Additionally, we have 4 Centre-Forwards, 3 Left-Wingers, 1 Right-Winger, 1 Attacking Midfield, and 1 Center-Midfielder.
- Analyzing by club, we have two players from Manchester City and two from Paris Saint-Germain among the top 10 most valued players. The other clubs have one player each. The clubs are as follows: Borussia Dortmund, Tottenham Hotspur, Liverpool FC, Chelsea FC, Manchester United, and FC Barcelona.

Mbappé x Haaland



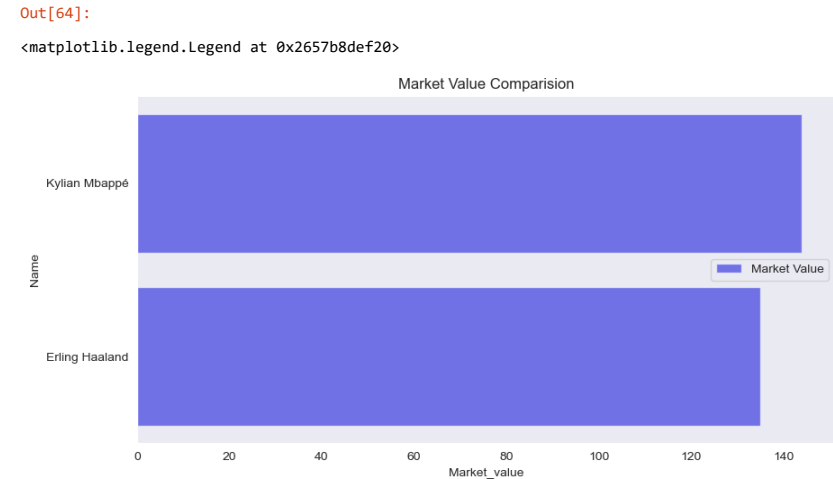
Lets analyze and compare the two most valuable players in the world.

```
In [63]:
top_2 = data[data['Name'].isin(['Erling Haaland', 'Kylian Mbappé'])]
top_2
```

Out[63]:

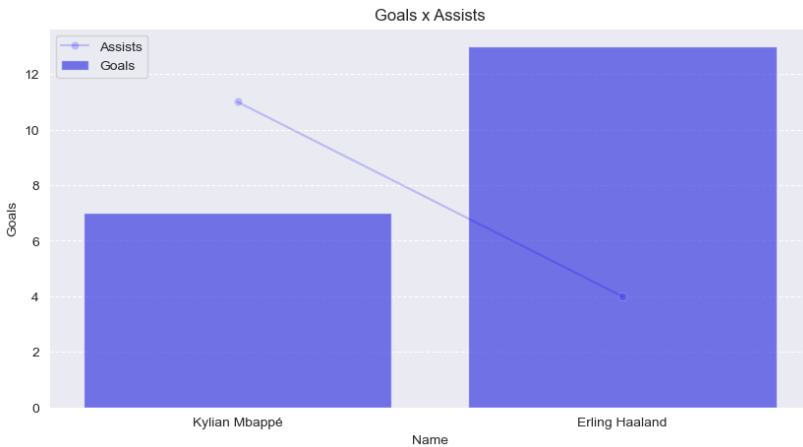
	Name	Position	Age	Market_value	Country	Club	Matches	Goals	Own_goals	Assists	Yellow_cards	Se
0	Kylian Mbappé	Centre-Forward	22	144	France	Paris Saint-Germain	16	7	0	11	3	
1	Erling Haaland	Centre-Forward	21	135	Norway	Borussia Dortmund	10	13	0	4	1	

```
In [64]:
# visualizing
plt.figure(figsize=(10,5))
plt.title("Market Value Comparision")
sns.set_style("darkgrid")
sns.barplot(y='Name', x='Market_value', data=top_2, color='blue', alpha=0.6, label='Market Value')
plt.legend()
```

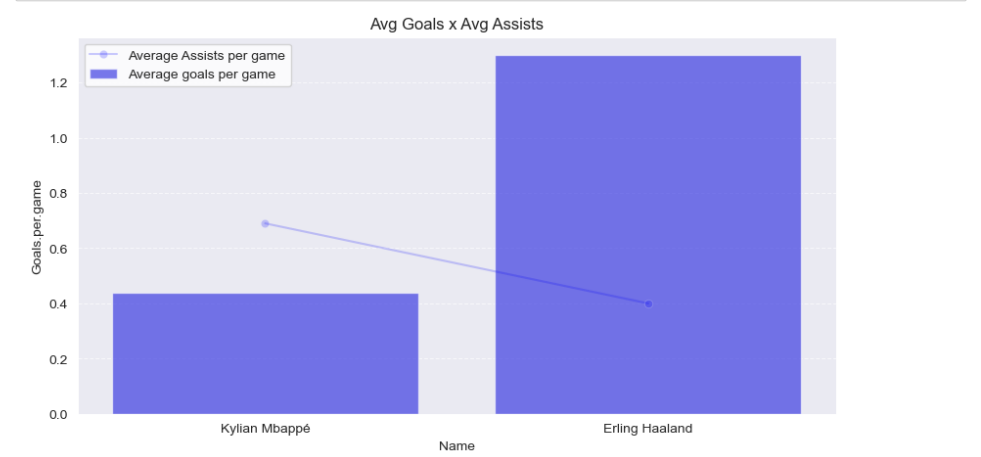


In [65]:

```
# visualizing
plt.figure(figsize=(10,5))
plt.title("Goals x Assists")
sns.barplot(x='Name', y='Goals', data=top_2, color='blue', alpha=0.6, label='Goals')
sns.lineplot(data=top_2['Assists'], color='blue', alpha=0.2, marker='o', label='Assists')
plt.grid(True, axis='y', linestyle='--', alpha=1.0)
sns.set_style("dark")
plt.legend()
plt.show()
```



```
In [66]:  
  
# visualizing  
  
plt.figure(figsize=(10,5))  
  
plt.title("Avg Goals x Avg Assists")  
  
sns.barplot(x='Name', y='Goals.per.game', data=top_2, color='blue', alpha=0.6, label='Average goals per game')  
sns.lineplot(data=top_2['Assists.per.game'], color='blue', alpha=0.2, marker='o', label='Average Assists per game')  
  
plt.grid(True, axis='y', linestyle='--', alpha=0.7)  
  
sns.set_style("white")  
  
plt.legend()  
  
plt.show()
```



Insights

Portuguese

- Percebemos que Kylian Mbappé está na frente de Erling Haaland em relação a valor de mercado.
- Quando comparamos gols e assistências, percebemos duas dinâmicas opostas: Mbappé apresentou maior número de assistências. Por outro lado, Haaland teve maior número de gols no ano. Essa mesma tendência se repetiu para média de gols por partida e média de assistências por partida.

English

- We noticed that Kylian Mbappé is ahead of Erling Haaland in terms of market value.
- When comparing goals and assists, we observed two opposite dynamics: Mbappé had more assists, while Haaland scored more goals. This same trend repeated for the average goals per match and average assists per match.